Hotel Reservation System

Flowcharts with predefined processes

Object-oriented hotel reservation system built from hotel-core (shared business rules), hotel-desktop (Java Swing, staff) and hotel-web (Spring Boot + Thymeleaf, customers). Both apps use one shared database. These flowcharts cover every process in the project guide and README.

How to read this set

- Start with the Whole System Flowchart (page 4). It shows the whole system in one view.
- A blue box with double side bars is a predefined process: a complex step that has its own page. The box says which page to open (for example, see P7).
- Every detail page starts at a Start terminator and ends at a Return terminator. Return means: go back to the box that called this page.
- Some detail pages call other detail pages. The index on the next page lists where each one is called from.
- Boxes on the main flowchart use the method names from the code, such as manageRooms(). The detail page shows the same method name under its title.
- The customer website (hotel-web, final phase) appears as its own box on the main flowchart. P11 and P13 describe it.

Symbols

	**Terminator**	Start, End, or Return to the caller
	**Process**	One action done by the system or by staff
	**Predefined process**	A detailed process drawn on its own page
	**Decision**	A yes/no question. Each exit is labelled
	**Input / output**	A person enters data or sees a result
	**Database**	Data saved to, or read from, the shared database
	**Document**	A receipt or a worked example
	**On-page connector**	Jump to the matching letter on the same page
	**Error or blocked**	Validation failed or the action is not allowed
	**Dashed two-way arrow**	Reads or writes data, not a step in the flow

Typical guest journey

Walk-in: P5 Create Walk-in Reservation → P14 Check In Guest → P15 Check Out Guest
Online: P11 Submit Online Reservation (website) → P12 Review Online Reservation (staff) → P14 Check In Guest → P15 Check Out Guest

Index of predefined processes

Open the page number to see the detail. Page numbers match the bookmarks in your PDF viewer.

CODE	PROCESS	WHAT IT DOES	CALLED FROM	PAGE
Main	**Whole System Flowchart**	Staff desktop application, plus the customer website (hotel-web). Each blue box with double side bars is drawn in detail on the page named inside the box.	-	**4**
P1	**Manage Rooms** `manageRooms()`	Pick one of three room tasks. Each task is its own predefined process, and Add and Update share the same validation (P2).	Main	**5**
P2	**Validate Room Data**	Rules applied in the core, so a desktop form cannot bypass them.	P19, P20	**6**
P3	**Update Housekeeping Status** `updateHousekeepingStatus()`	Staff sets a room to Clean, Dirty, Under Maintenance or Out of Service. A room with a checked-in guest cannot be taken out of service.	Main, P15	**7**
P4	**Monitor Rooms** `roomMonitoring()`	Read-only room dashboard for the front desk.	Main	**8**
P5	**Create Walk-in Reservation** `walkInReservation()`	Receptionist builds a booking at the desk. Each step is a predefined process that can reject and loop.	Main	**9**
P6	**Enter and Validate Guest Details** `validateGuestDetails()`	Same rules for the desktop form and the website form.	P5, P11	**10**
P7	**Verify Discount ID** `verifyDiscountID()`	A discounted guest gets 20% off and no tax, so eligibility is checked and recorded.	P5, P22	**11**
P8	**Enter and Validate Stay Dates** `validateGuestStayDates()`	Nights are always calculated from the dates, never taken from the form or browser.	P5, P11	**12**
P9	**Check Room Availability** `checkRoomAvail()`	Checks the whole stay, not just today's occupancy label.	P5, P11, P12, P23	**13**
P10	**Save Reservation** `saveReservation()`	Locks the room, checks again, then saves in one transaction. Stops two users booking the same room for overlapping dates.	P5, P11	**14**
P11	**Submit Online Reservation** `submitOnlineReservation()`	Customer-facing flow on the website. The booking is saved as Online + Pending and no card payment is taken.	Main (hotel-web)	**15**
P12	**Review Online Reservation** `onlineReservation()`	Staff confirm or cancel Pending bookings in the desktop app. A Confirmed booking can be moved to a replacement room.	Main	**16**
P13	**Find or Cancel Reservation** `findOrCancelReservation()`	Optional website page. Never reveals another guest's information.	Main (hotel-web)	**17**
P14	**Check In Guest** `checkInGuest()`	Works for walk-in and online guests. Online guests are classified here, using their ID at the desk.	Main, P5	**18**
P15	**Check Out Guest** `checkOutGuest()`	Ends the stay: bill, cash payment, receipt, then the room goes Dirty until cleaned.	Main	**19**
P16	**Calculate Bill**	Only the room is billed. Use BigDecimal for every money value.	P15	**20**
P17	**Process Cash Payment**	Cash only. Too little cash is rejected before anything is saved.	P15	**21**

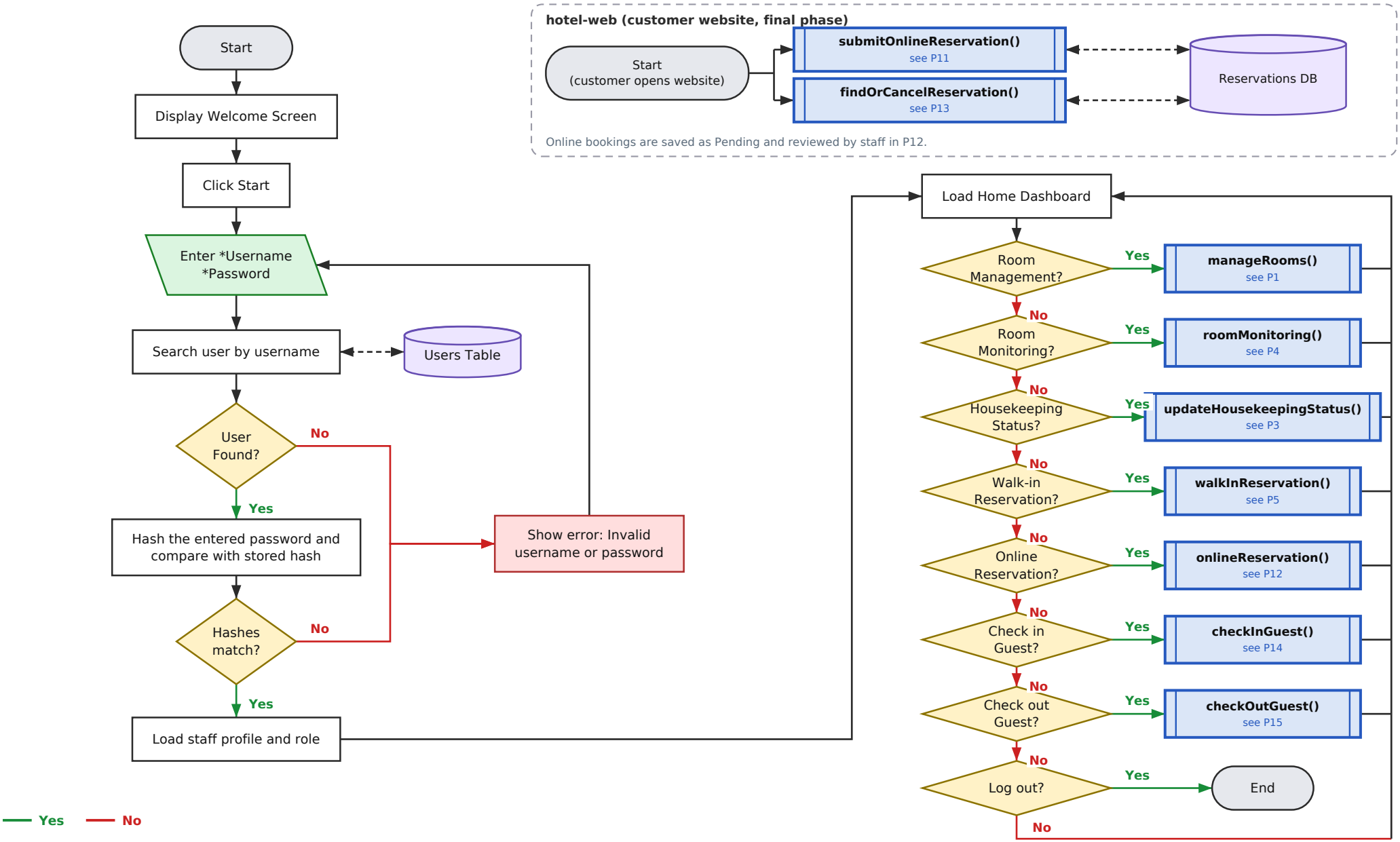
CODE	PROCESS	WHAT IT DOES	CALLED FROM	PAGE
P18	**Generate Receipt**	Receipt shown after a successful payment.	P15	**22**
P19	**Add Room**	Creates a room record with Available and Clean as the starting status.	P1	**23**
P20	**Update Room**	The room number identifies the record and cannot be changed.	P1	**24**
P21	**Search / Display Room**	Search by room number or room type.	P1	**25**
P22	**Confirm Discount Eligibility**	Online guests never state a classification on the website, so this is where it is settled.	P14	**26**
P23	**Ensure Room Is Ready**	A guest can only be checked in to a room that is Clean, not occupied and in service.	P14	**27**

Decisions made where the project documents are silent

- Walk-in reservations are saved as Confirmed. Online reservations are saved as Pending, as the README states.
- A guest can cancel a reservation only while it is Pending or Confirmed.
- Check-in is blocked before the arrival date and after the check-out date.
- A room with a checked-in guest cannot be set to Under Maintenance or Out of Service (P3).
- Online guests give no classification on the website, so discount eligibility is settled at check-in (P22).
- The discount ID check (name matches, ID not expired) is an assumed desk procedure (P7). Adjust it to your guide.
- If a Pending online booking cannot be confirmed in its room, staff look for a replacement room (P9); if none is found they contact the guest and the booking is Cancelled and its dates released (P12).
- Amounts are shown without a currency symbol because the documents do not name one.

Whole System Flowchart

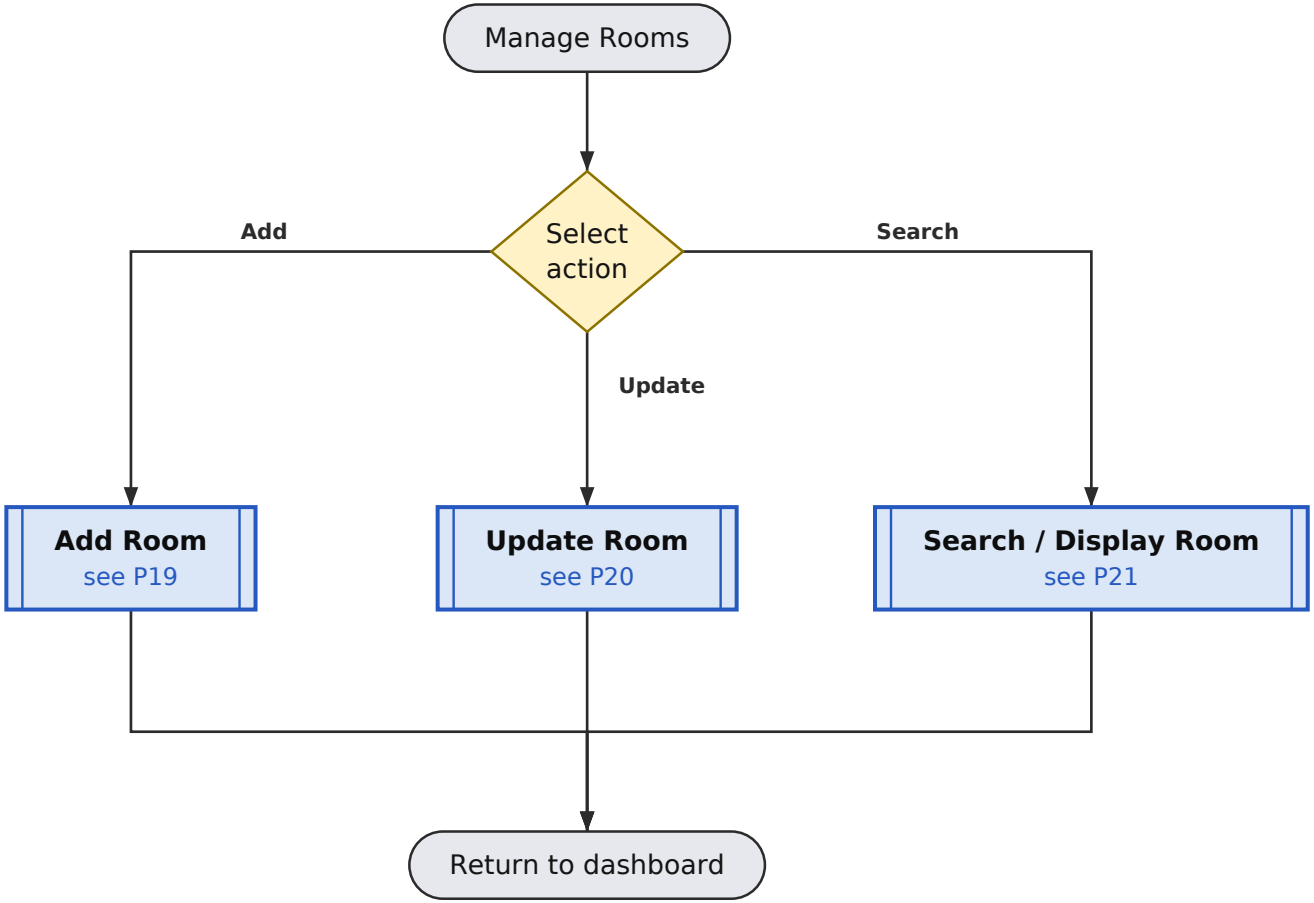
Staff desktop application, plus the customer website (hotel-web). Each blue box with double side bars is a predefined process, drawn in detail on the page named inside the box.



Blue boxes with double side bars are predefined processes. Open the page named in each box (see Pn).

P1 Manage Rooms `manageRooms()`

Pick one of three room tasks. Each task is its own predefined process, and Add and Update share the same validation (P2).

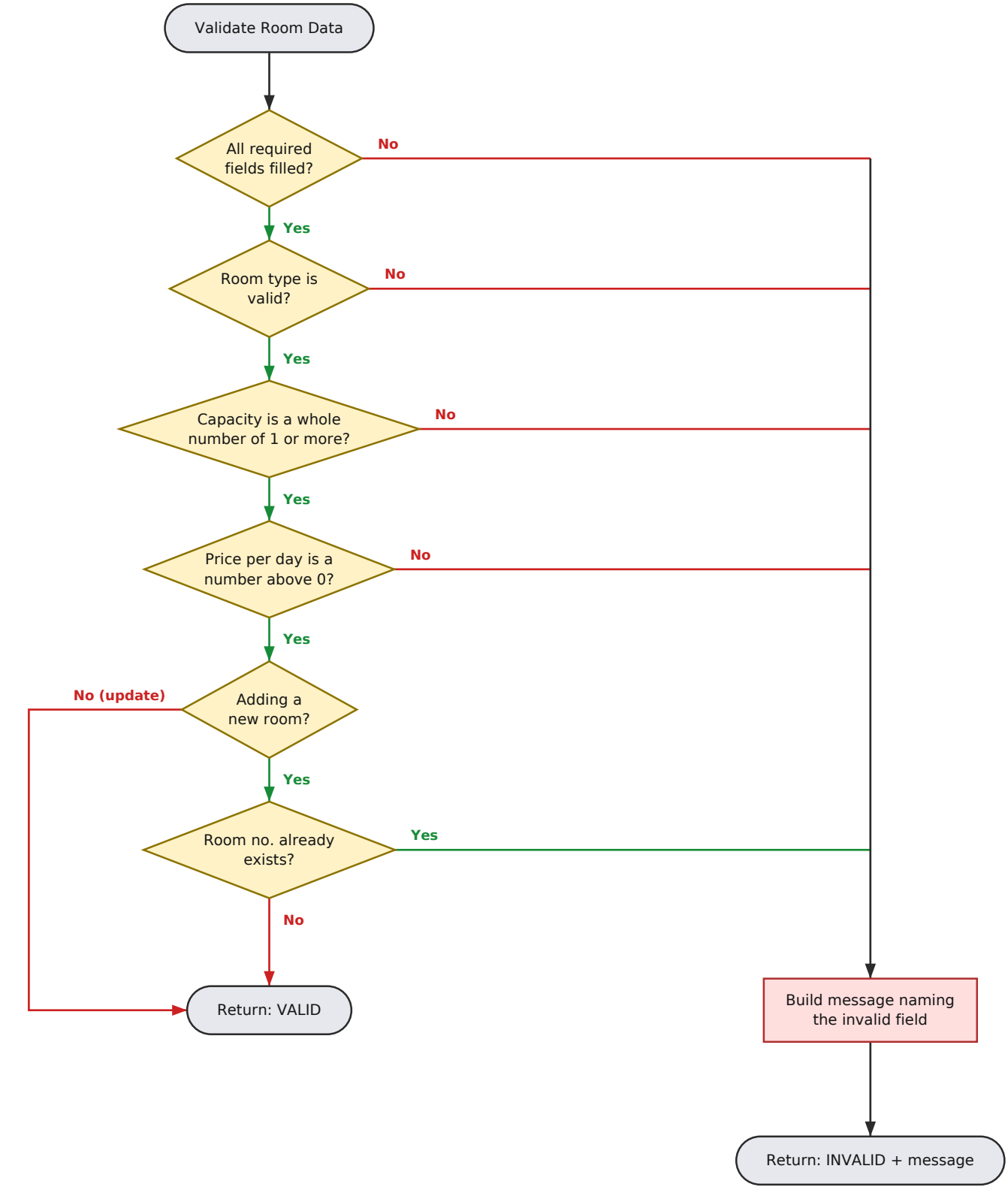


Yes No

Called from: Main. Return goes back to the calling box.

P2 Validate Room Data

Rules applied in the core, so a desktop form cannot bypass them.

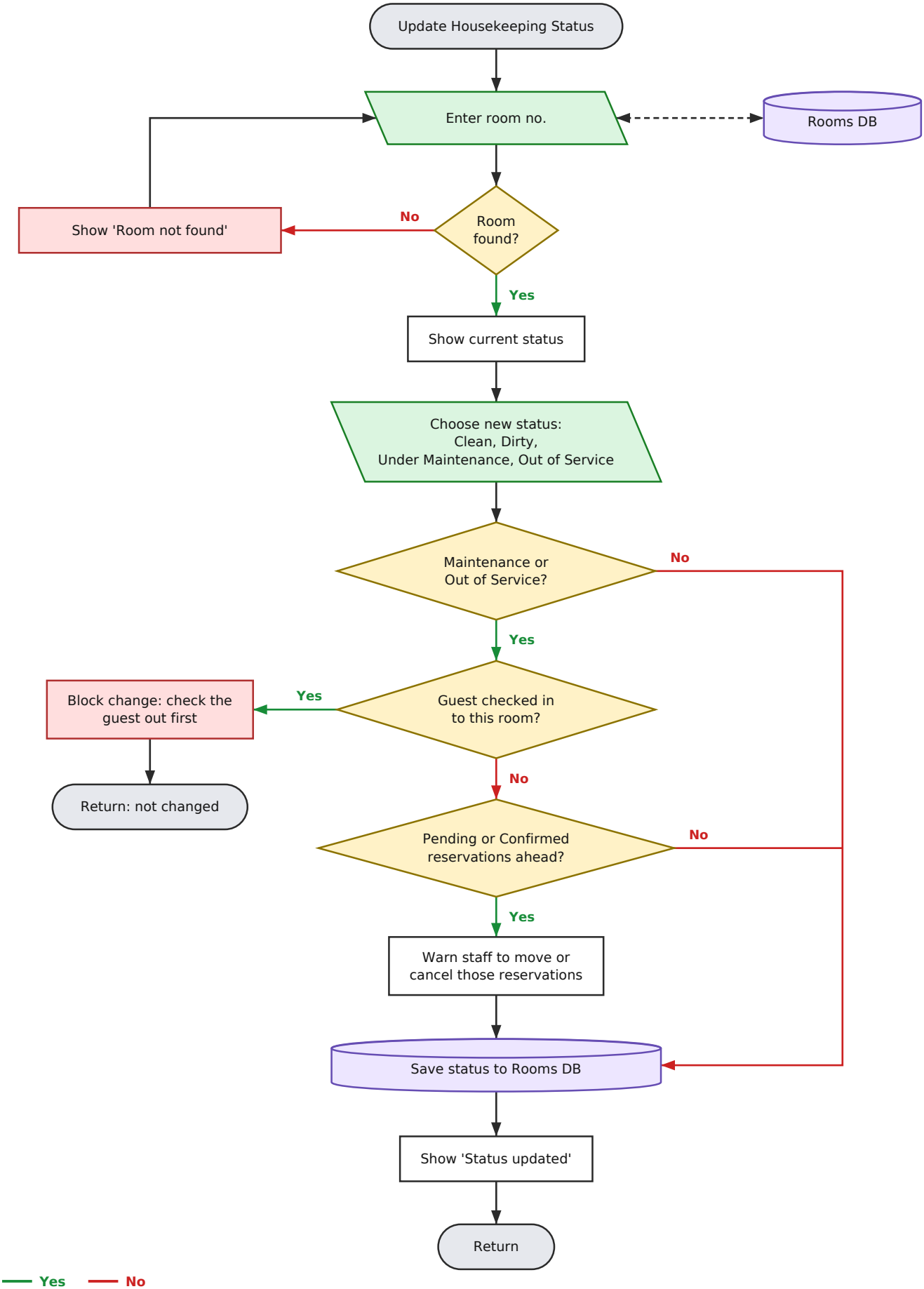


Yes No

Called from: P19, P20. Return goes back to the calling box.

P3 Update Housekeeping Status `updateHousekeepingStatus()`

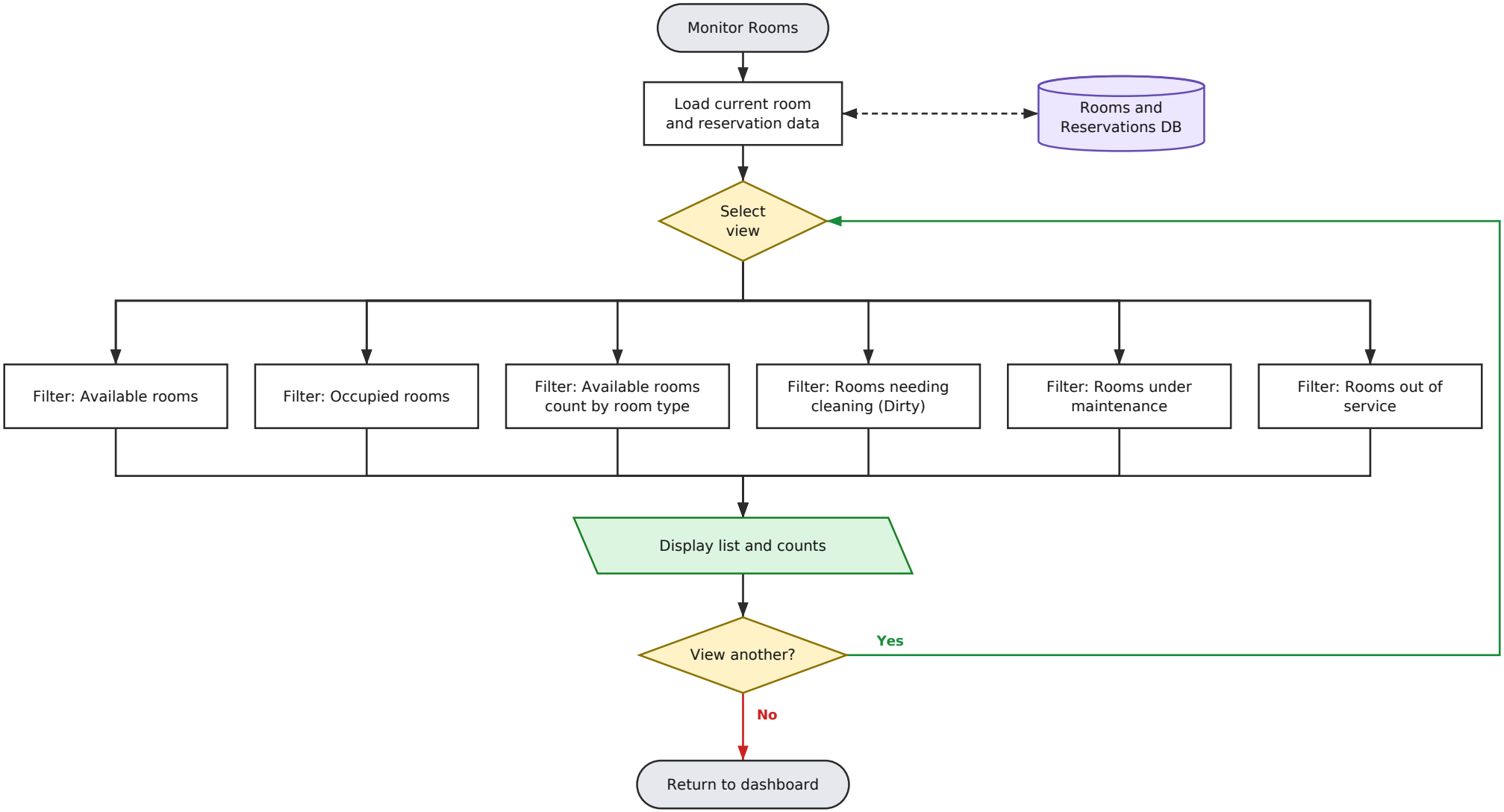
Staff sets a room to Clean, Dirty, Under Maintenance or Out of Service. A room with a checked-in guest cannot be taken out of service.



Called from: Main, P15. Return goes back to the calling box.

P4 Monitor Rooms `roomMonitoring()`

Read-only room dashboard for the front desk.

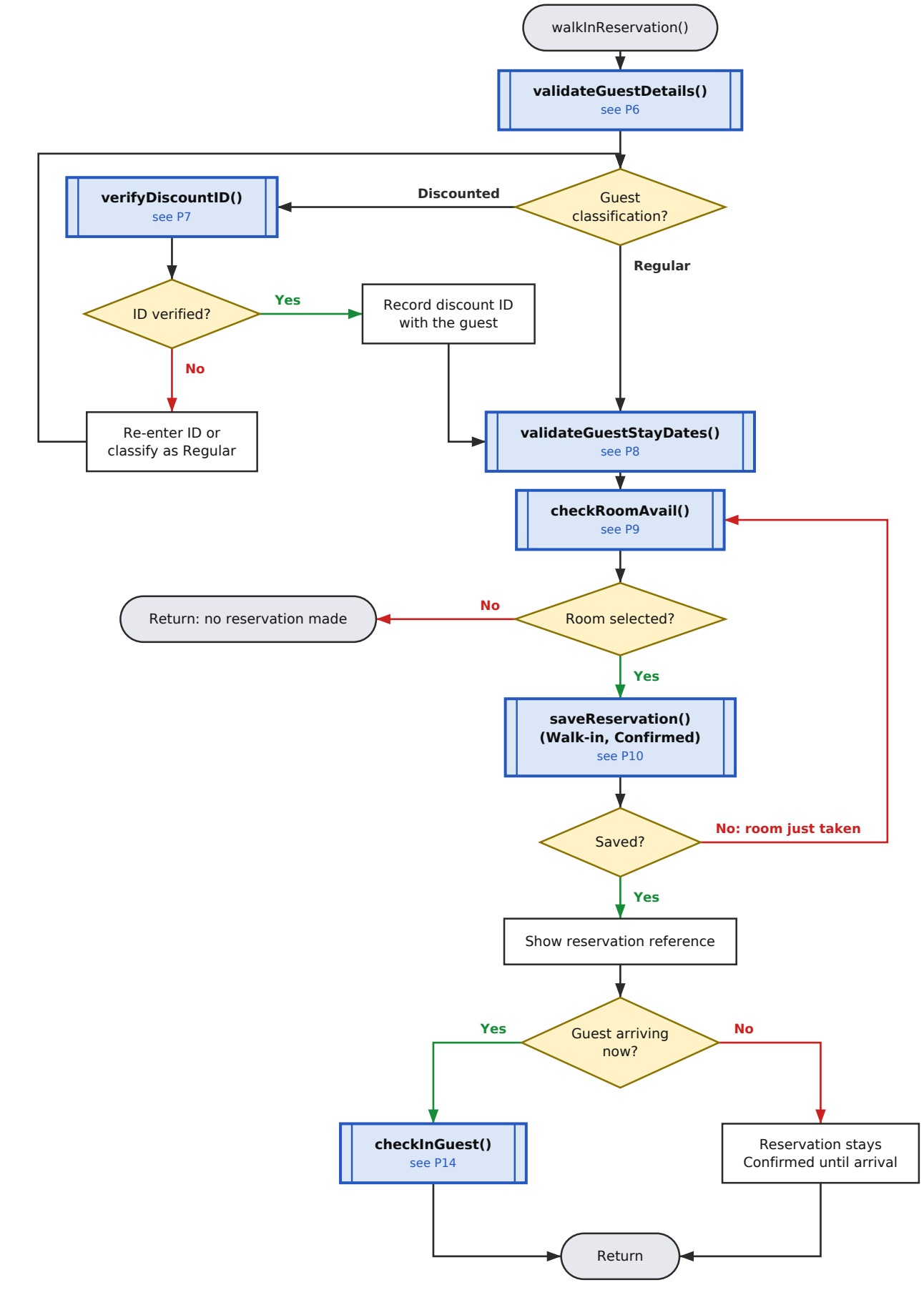


Yes No

Called from: Main. Return goes back to the calling box.

P5 Create Walk-in Reservation `walkInReservation()`

Receptionist builds a booking at the desk. Each step is a predefined process that can reject and loop.

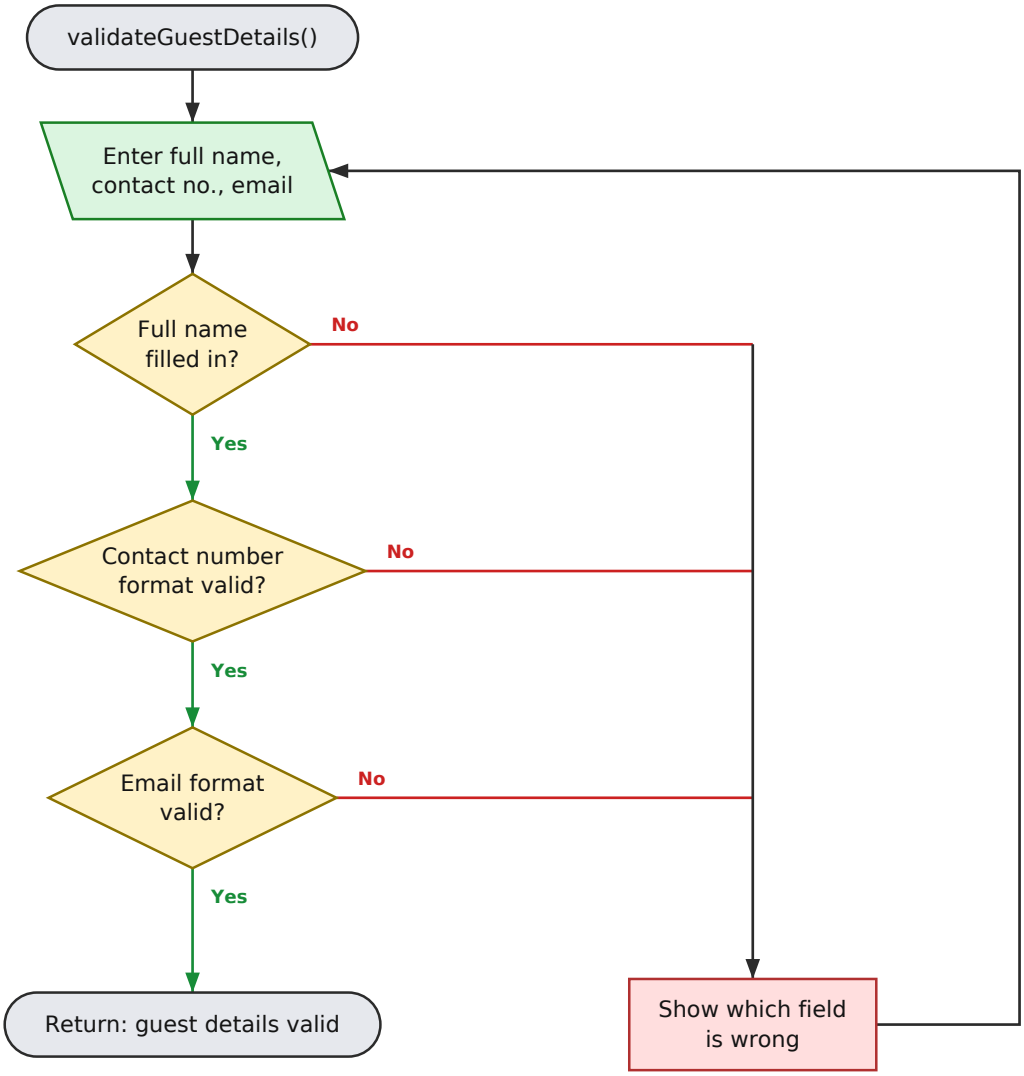


Yes No

Called from: Main. Return goes back to the calling box.

P6 Enter and Validate Guest Details `validateGuestDetails()`

Same rules for the desktop form and the website form.

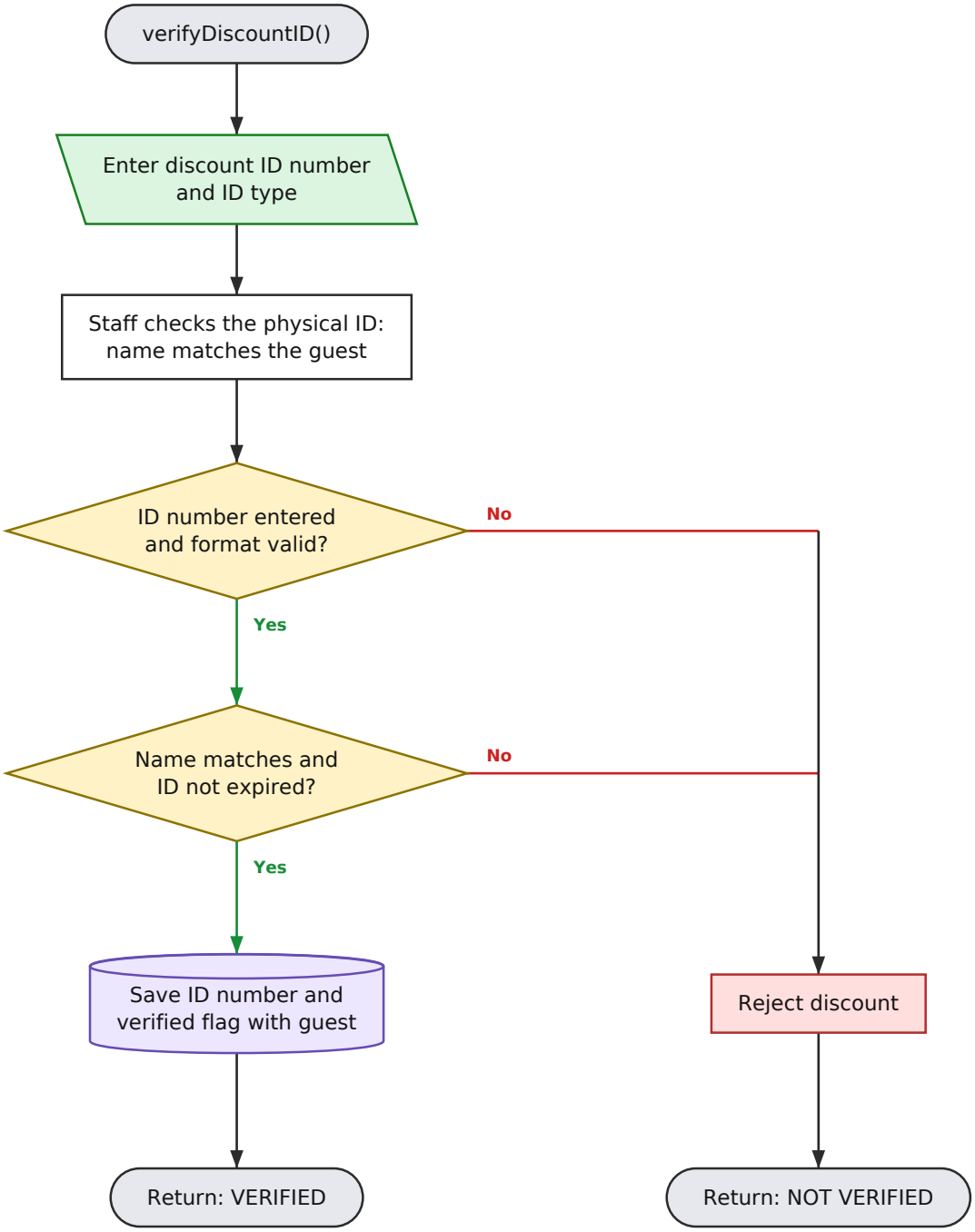


Yes No

Called from: P5, P11. Return goes back to the calling box.

P7 Verify Discount ID `verifyDiscountID()`

A discounted guest gets 20% off and no tax, so eligibility is checked and recorded.

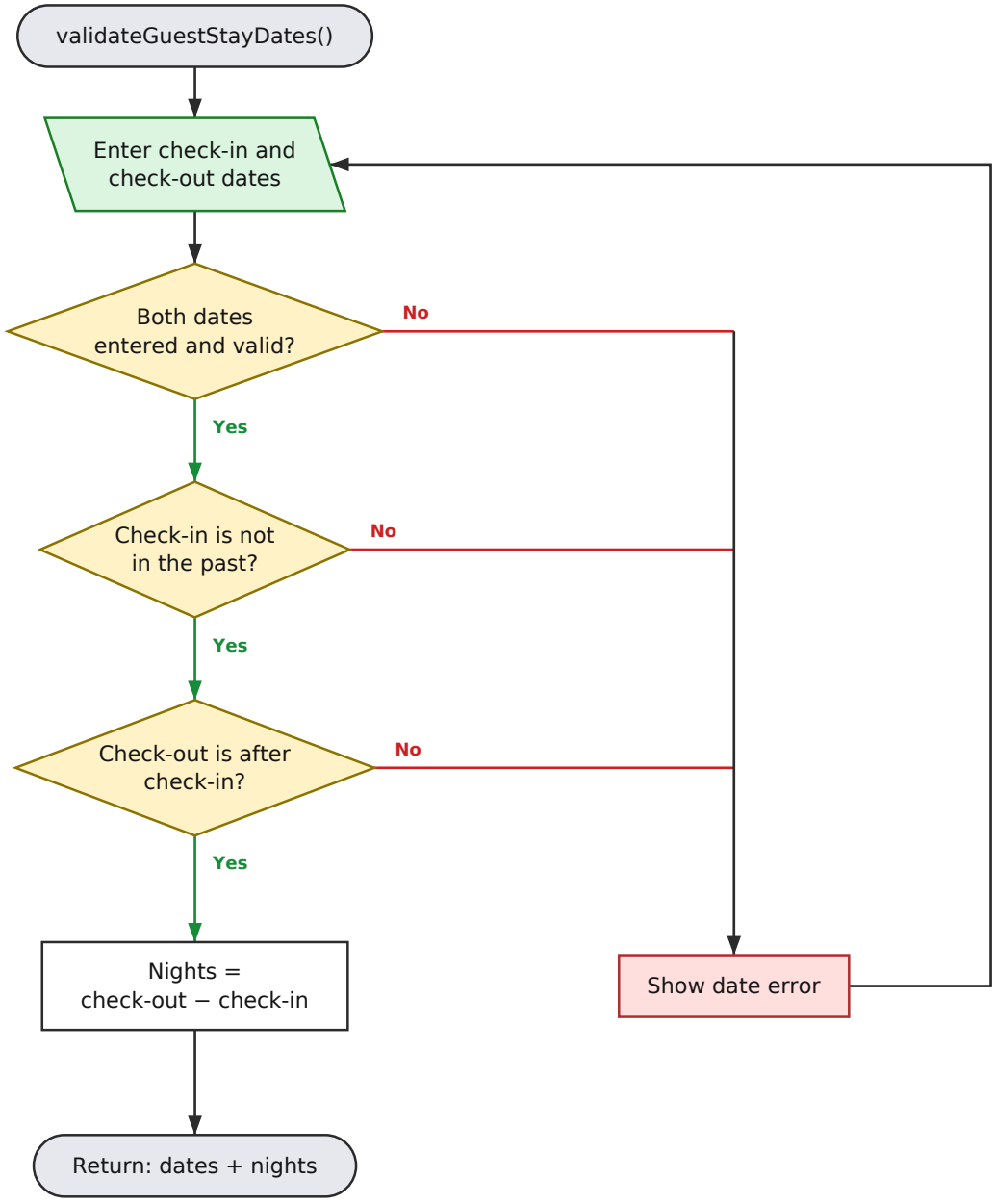


Yes No

Called from: P5, P22. Return goes back to the calling box.

P8 Enter and Validate Stay Dates `validateGuestStayDates()`

Nights are always calculated from the dates, never taken from the form or browser.

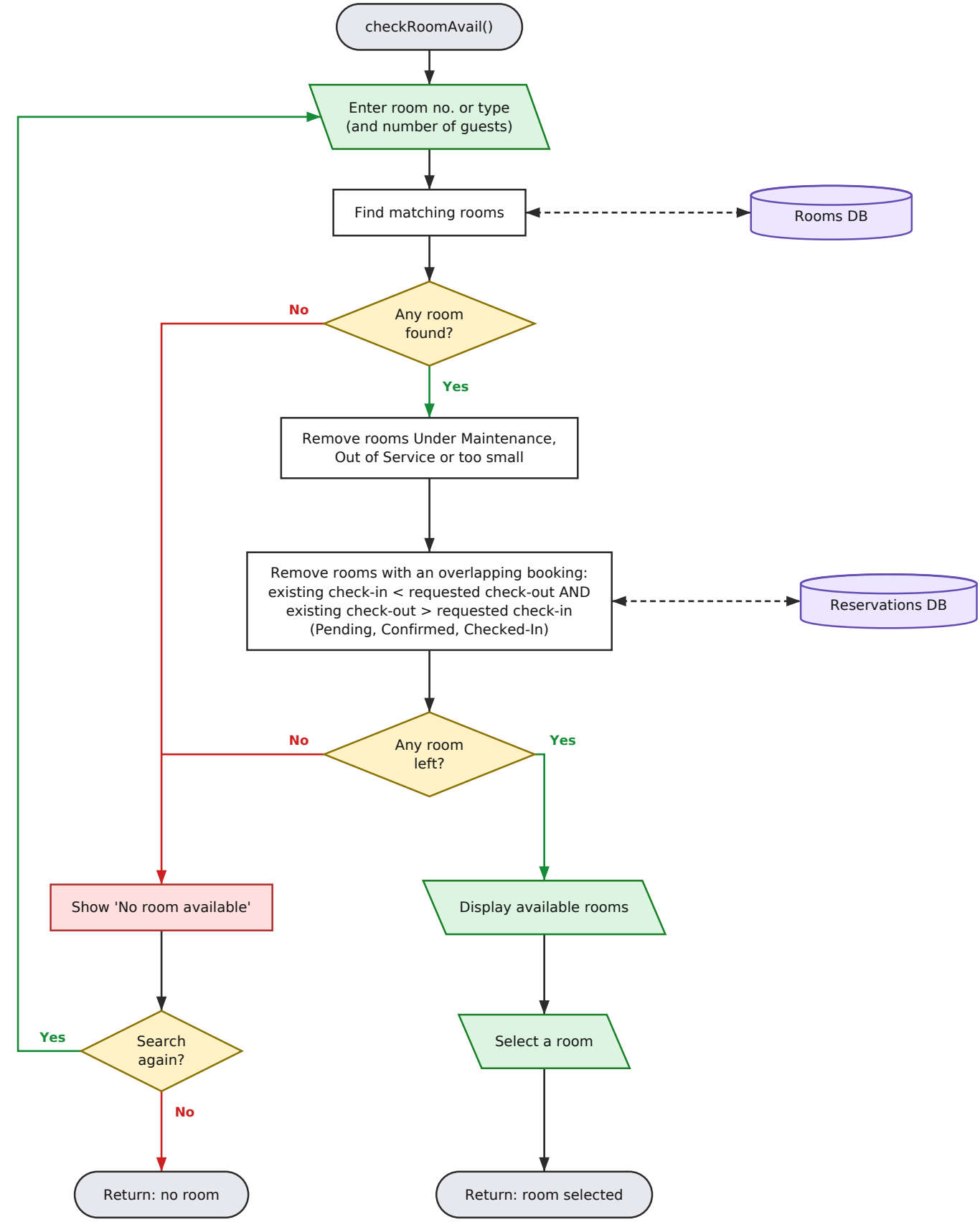


Yes No

Called from: P5, P11. Return goes back to the calling box.

P9 Check Room Availability `checkRoomAvail()`

Checks the whole stay, not just today's occupancy label.

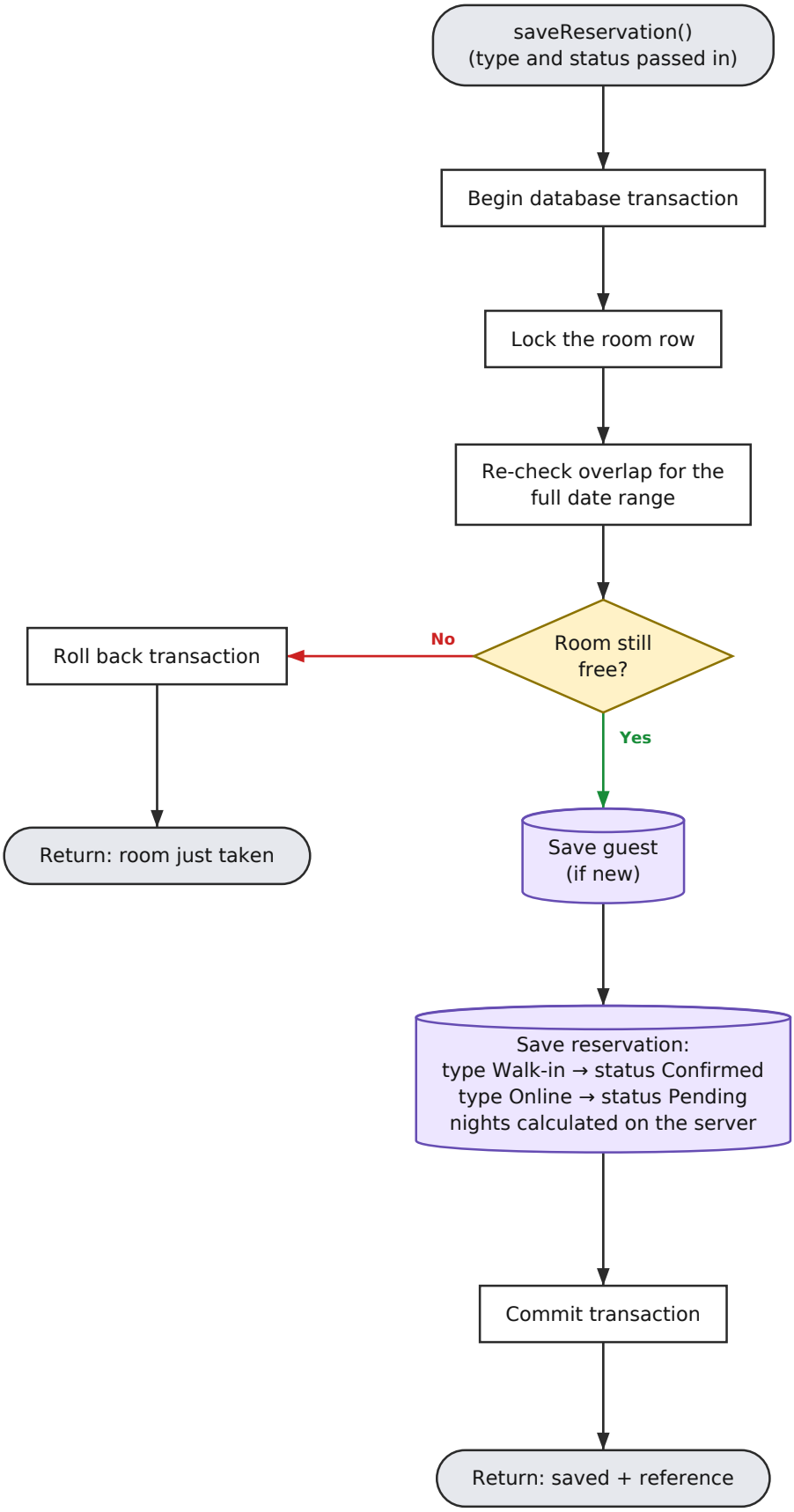


Yes No

Called from: P5, P11, P12, P23. Return goes back to the calling box.

P10 Save Reservation `saveReservation()`

Locks the room, checks again, then saves in one transaction. Stops two users booking the same room for overlapping dates.

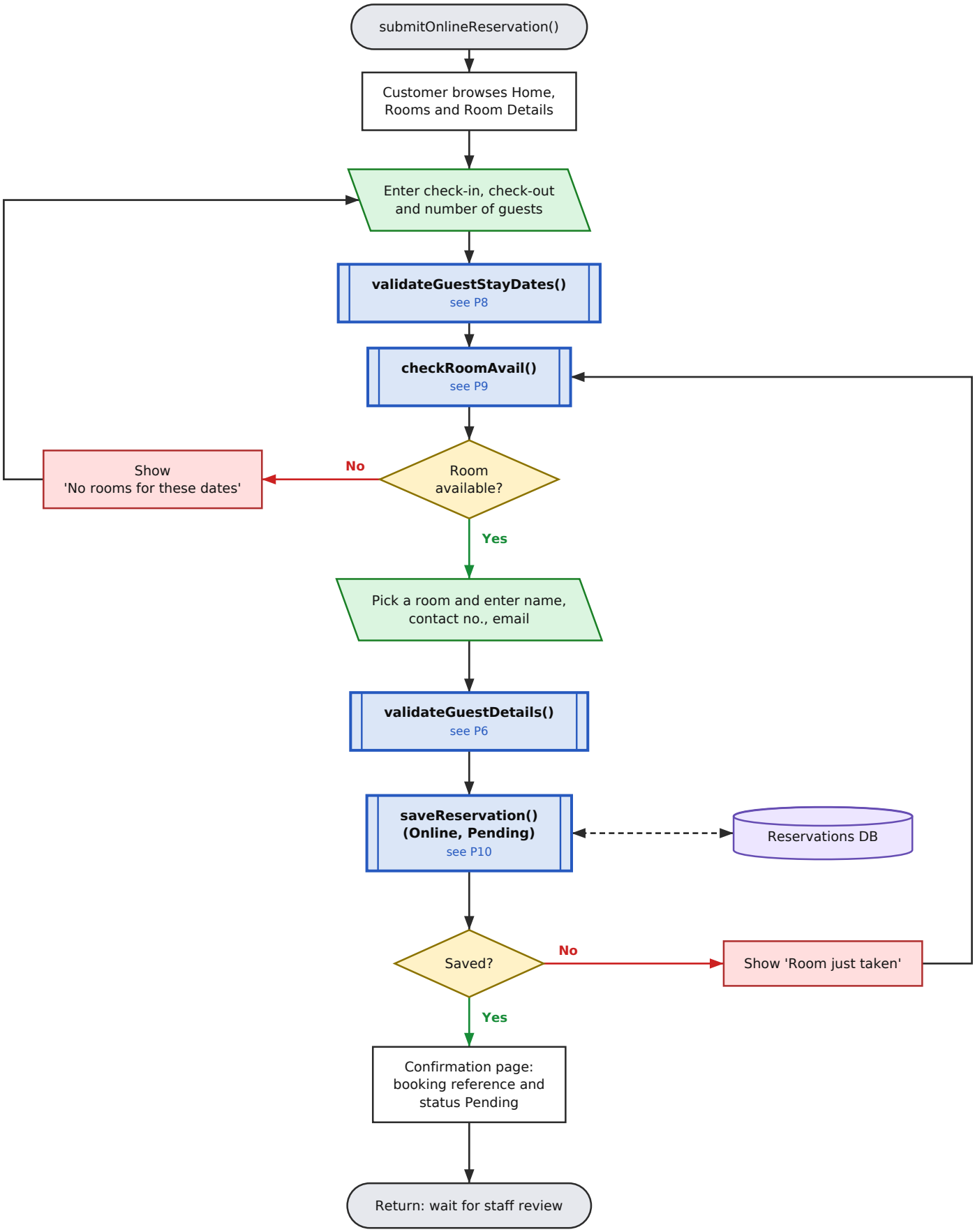


Yes No

Called from: P5, P11. Return goes back to the calling box.

P11 Submit Online Reservation `submitOnlineReservation()`

Customer-facing flow on the website. The booking is saved as Online + Pending and no card payment is taken.

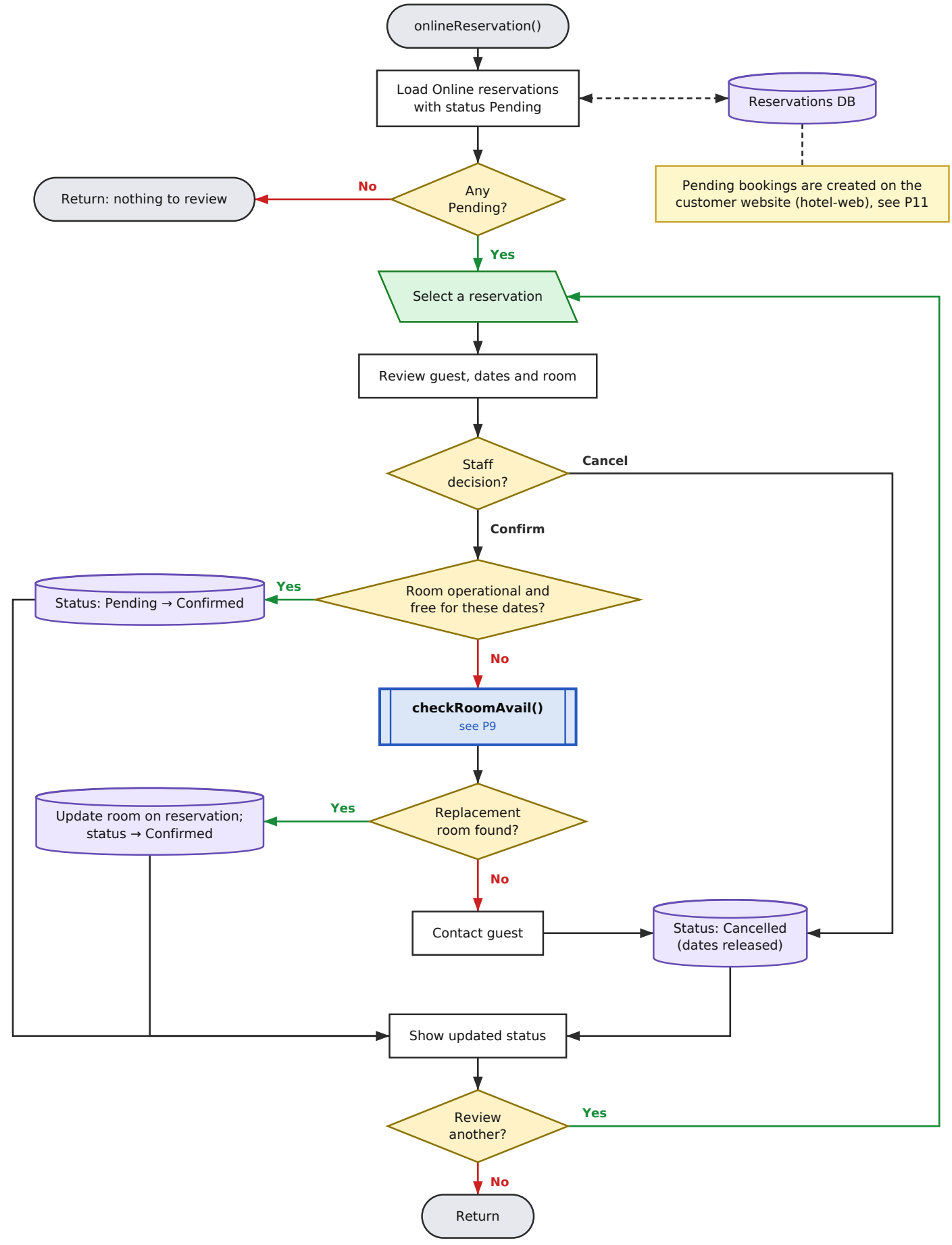


Yes No

Called from: Main (hotel-web box). Return goes back to the calling box.

P12 Review Online Reservation `onlineReservation()`

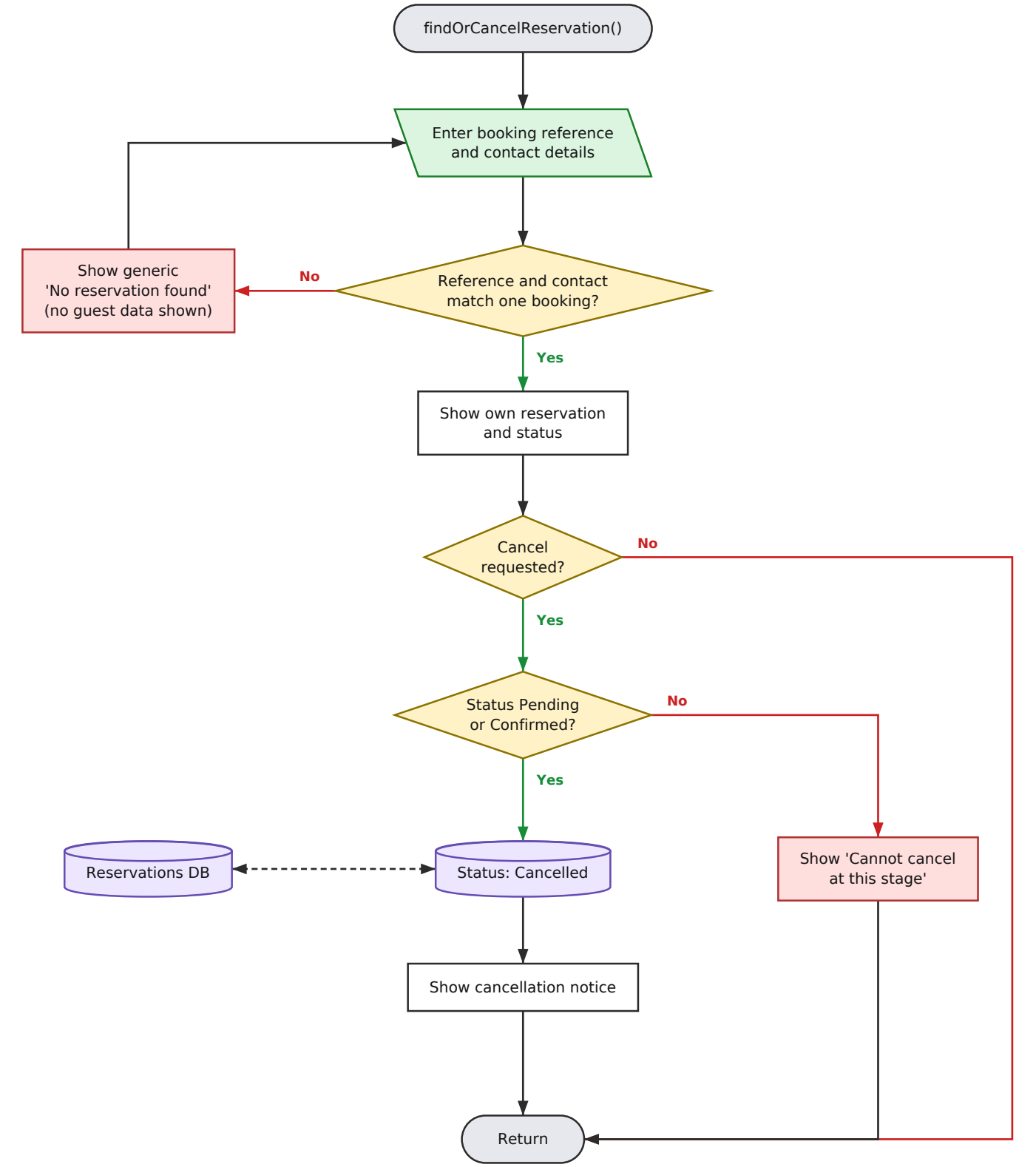
Staff confirm or cancel Pending bookings in the desktop app.



Called from: Main. Return goes back to the calling box.

P13 Find or Cancel Reservation `findOrCancelReservation()`

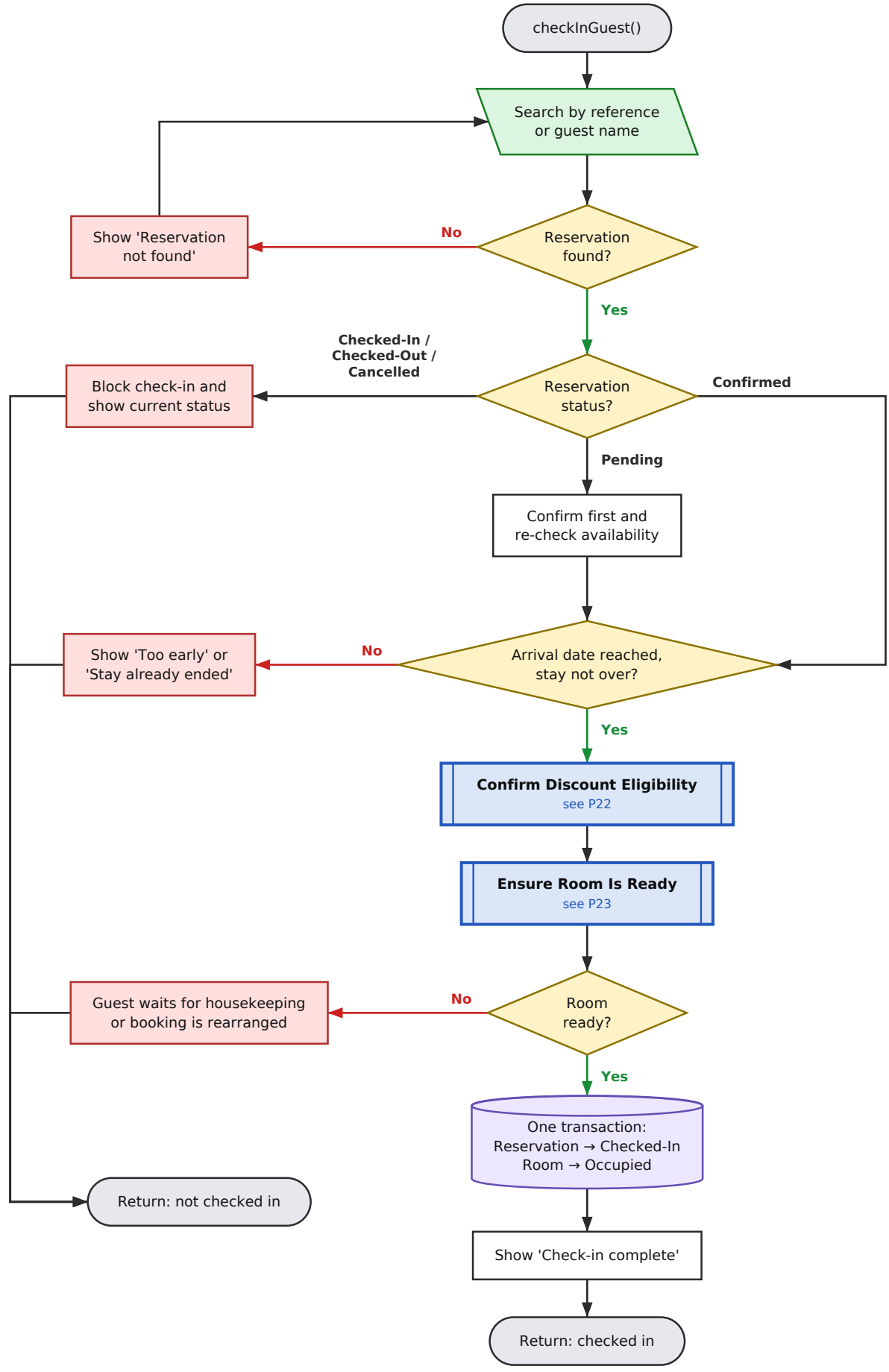
Optional website page. Never reveals another guest's information.



Yes No

P14 Check In Guest `checkInGuest()`

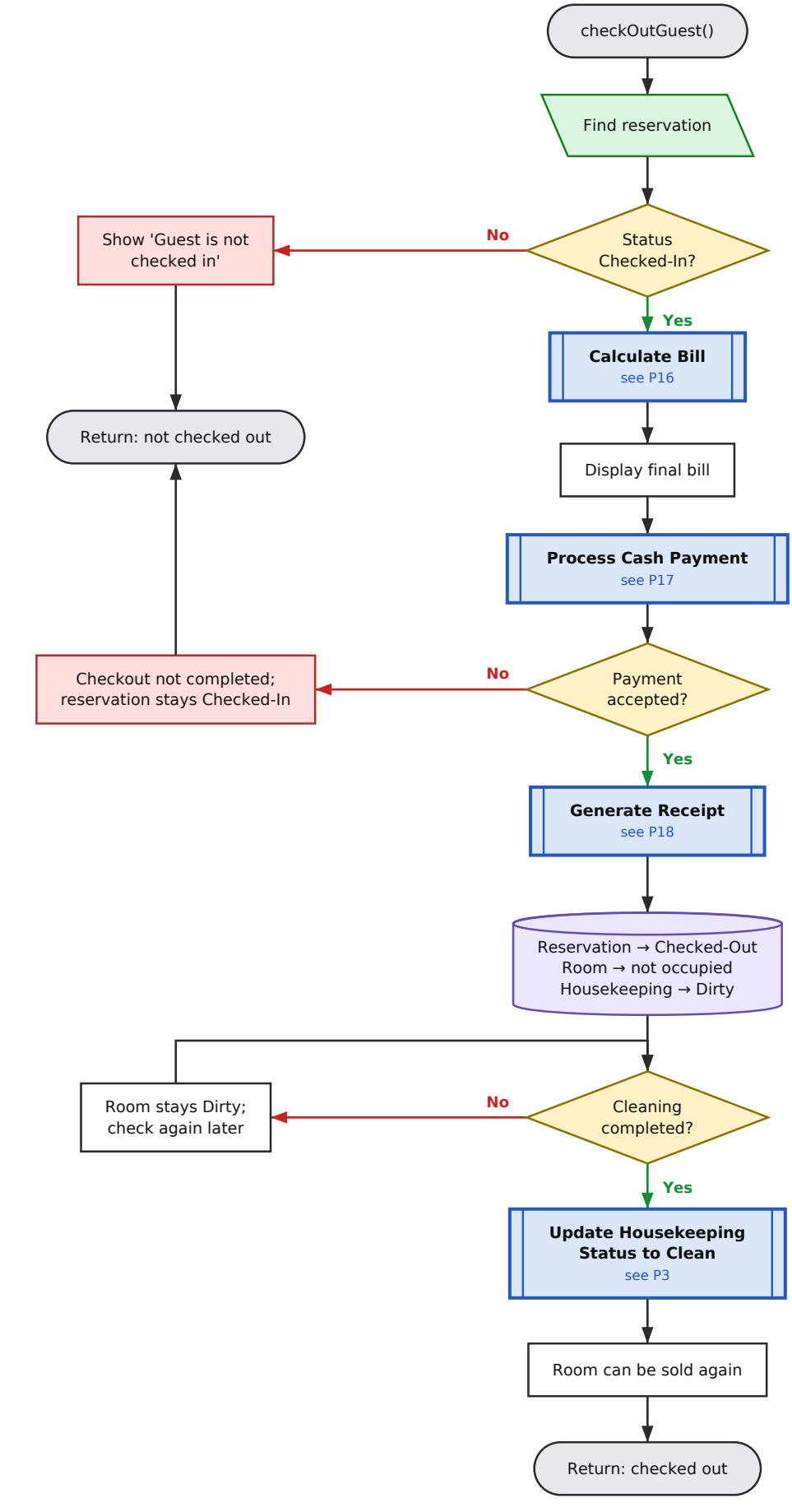
Works for walk-in and online guests. Online guests are classified here, using their ID at the desk.



Called from: Main, P5. Return goes back to the calling box.

P15 Check Out Guest `checkOutGuest()`

Ends the stay: bill, cash payment, receipt, then the room goes Dirty until cleaned.

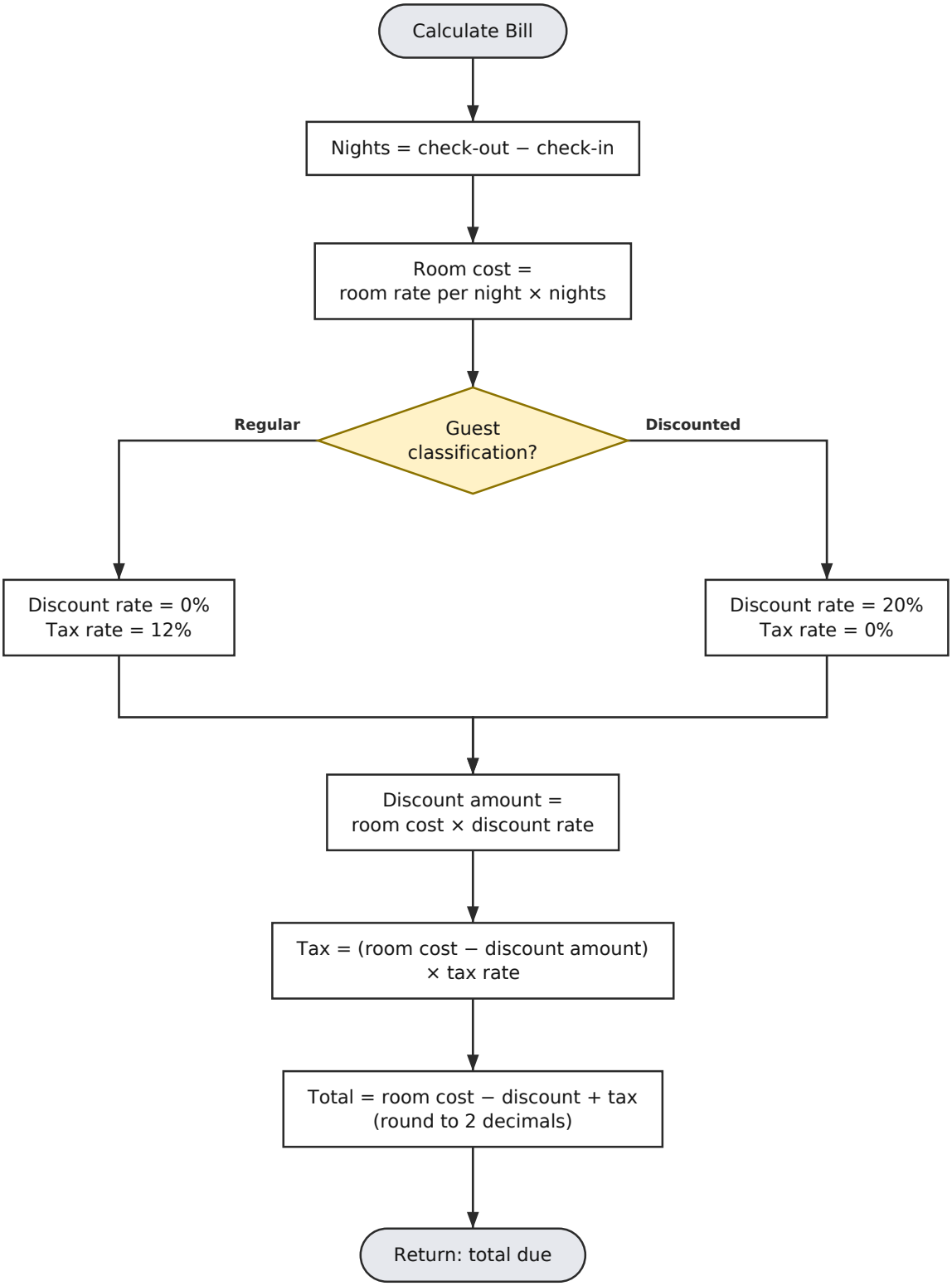


Yes No

Called from: Main. Return goes back to the calling box.

P16 Calculate Bill

Only the room is billed. Use BigDecimal for every money value.



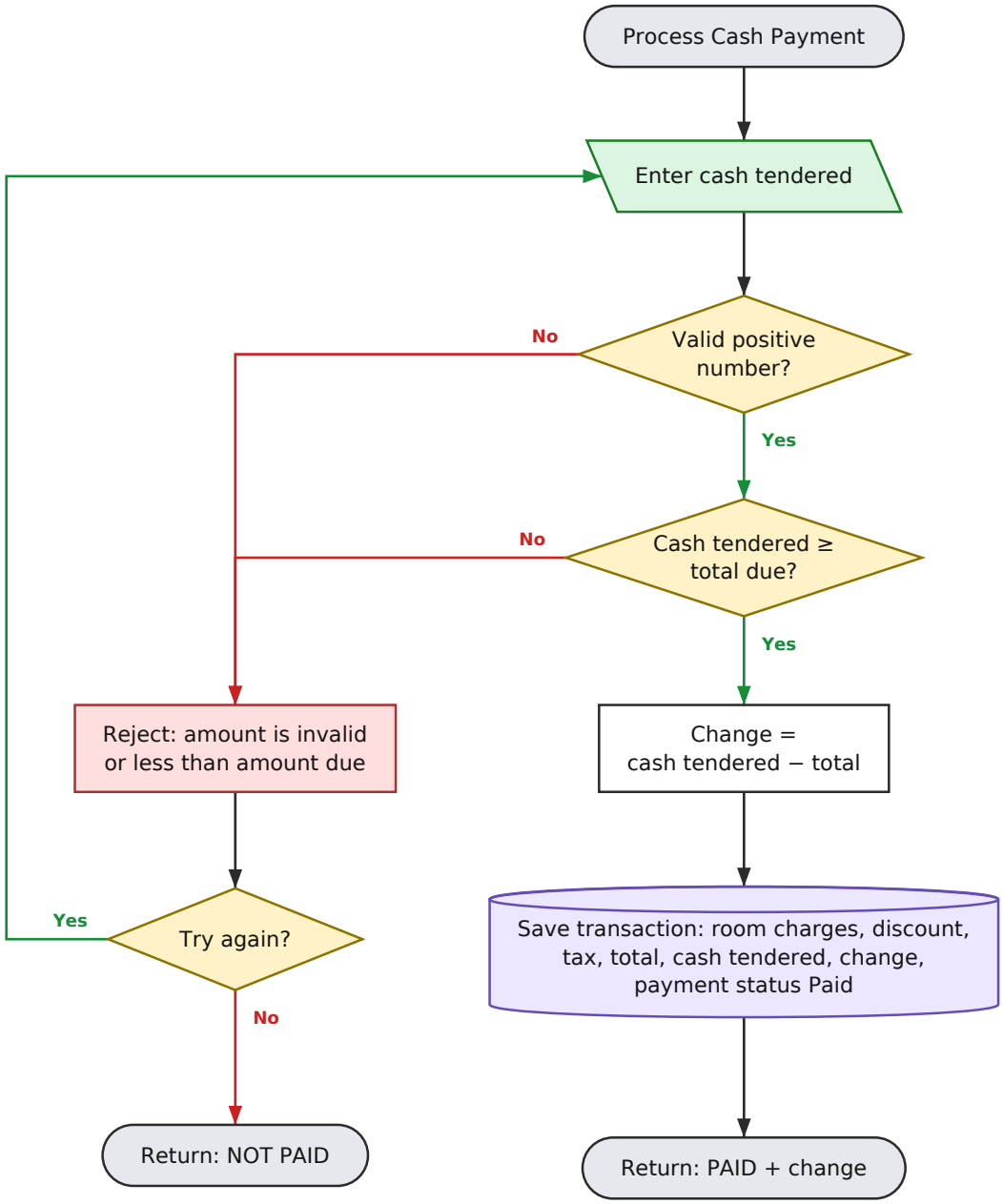
Example: rate 2,500.00 × 3 nights
Regular: 7,500.00 − 0.00 + 900.00 = 8,400.00
Discounted: 7,500.00 − 1,500.00 + 0.00 = 6,000.00

Yes No

Called from: P15. Return goes back to the calling box.

P17 Process Cash Payment

Cash only. Too little cash is rejected before anything is saved.

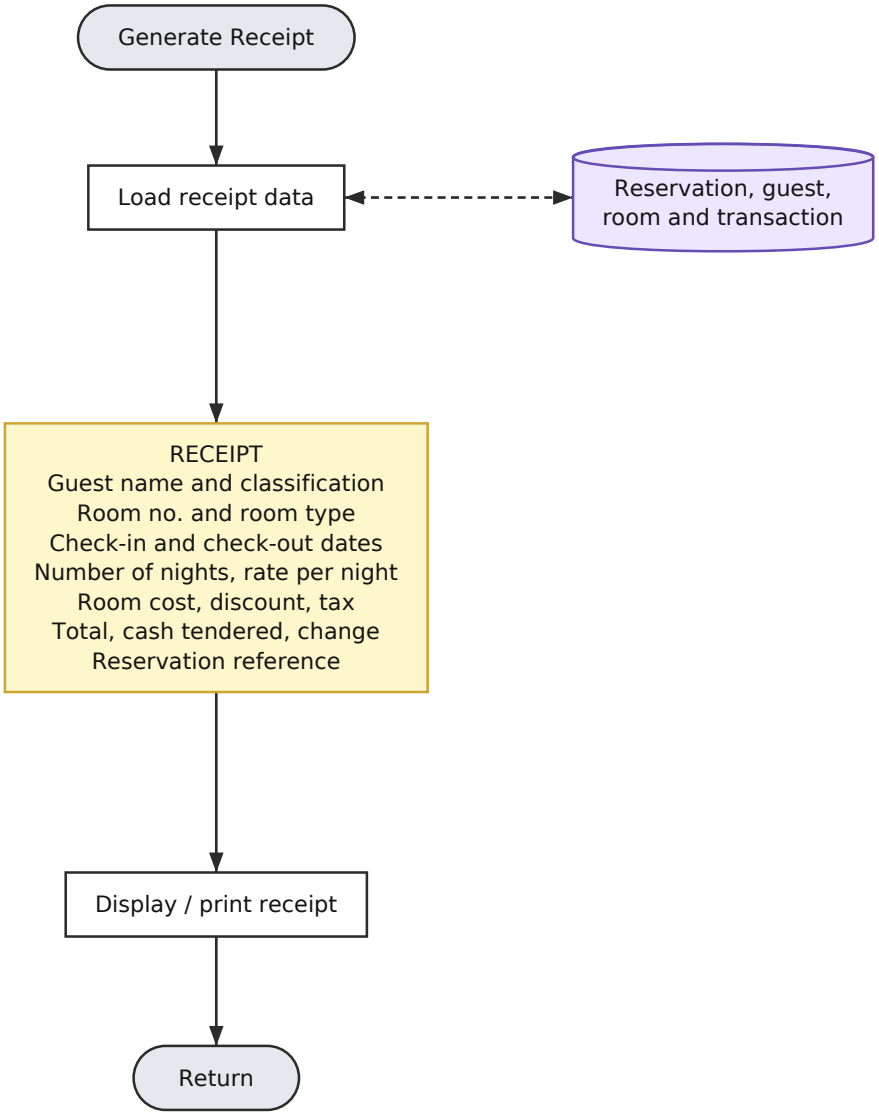


Yes No

Called from: P15. Return goes back to the calling box.

P18 Generate Receipt

Receipt shown after a successful payment.

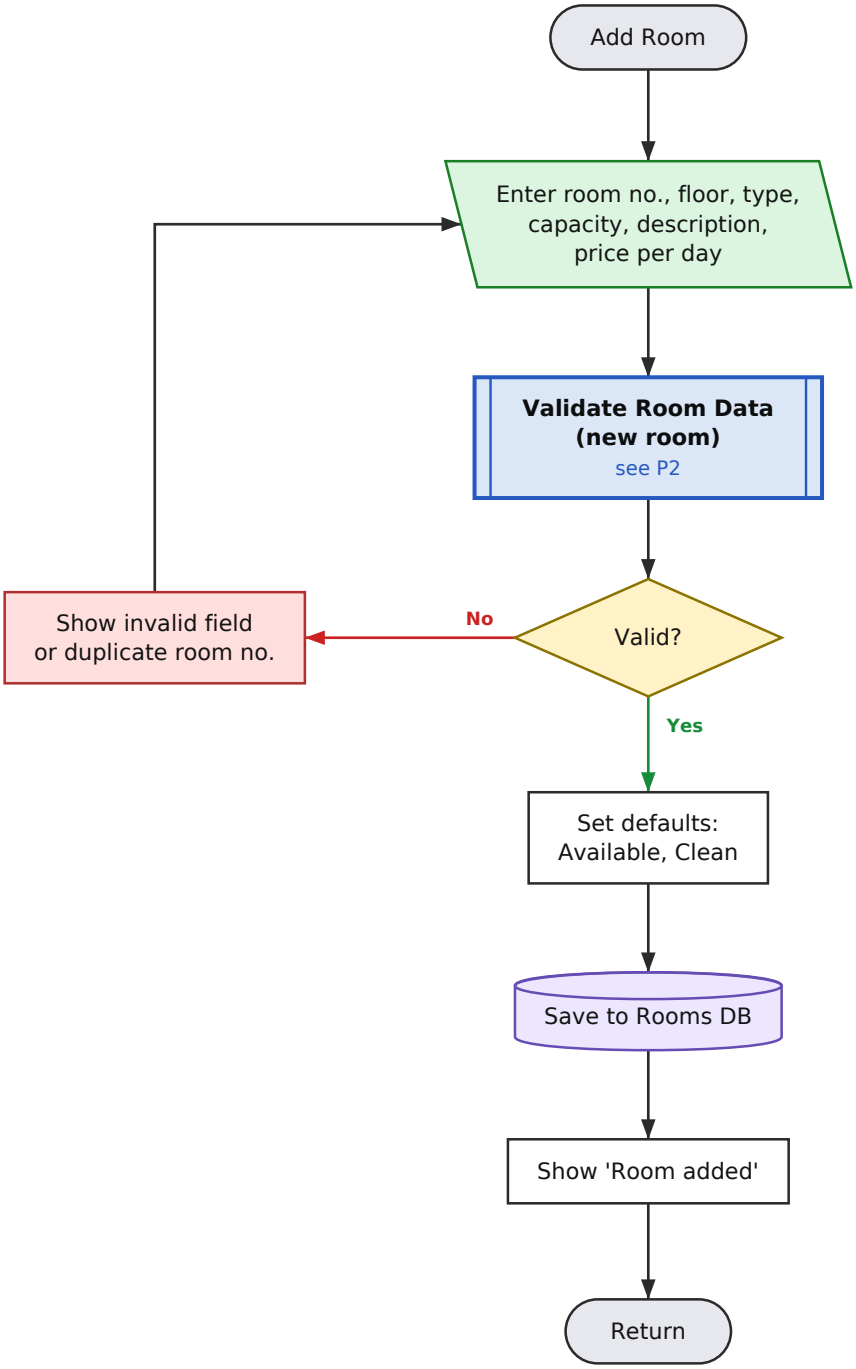


Yes No

Called from: P15. Return goes back to the calling box.

P19 Add Room

Creates a room record with Available and Clean as the starting status.

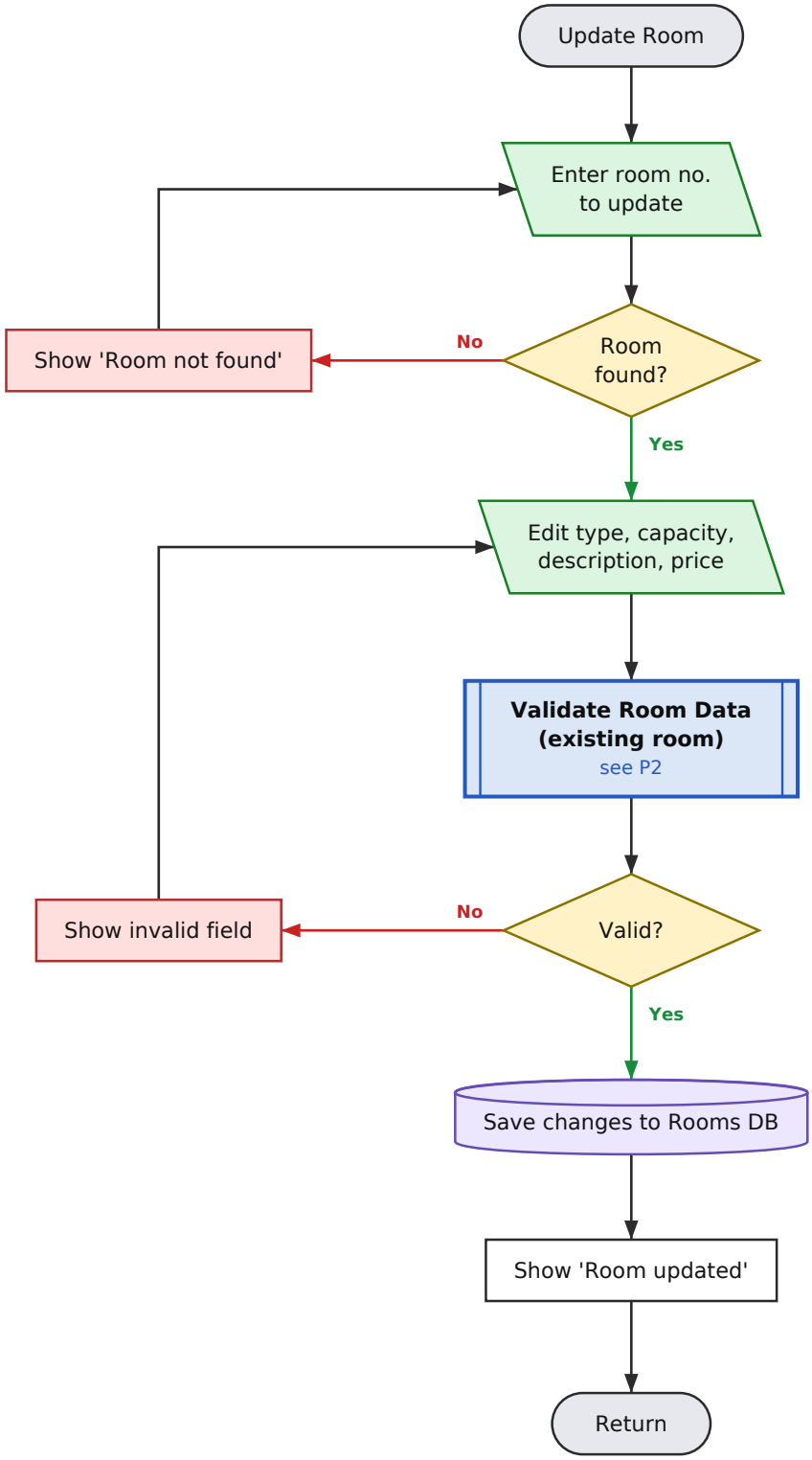


Yes No

Called from: P1. Return goes back to the calling box.

P20 Update Room

The room number identifies the record and cannot be changed.

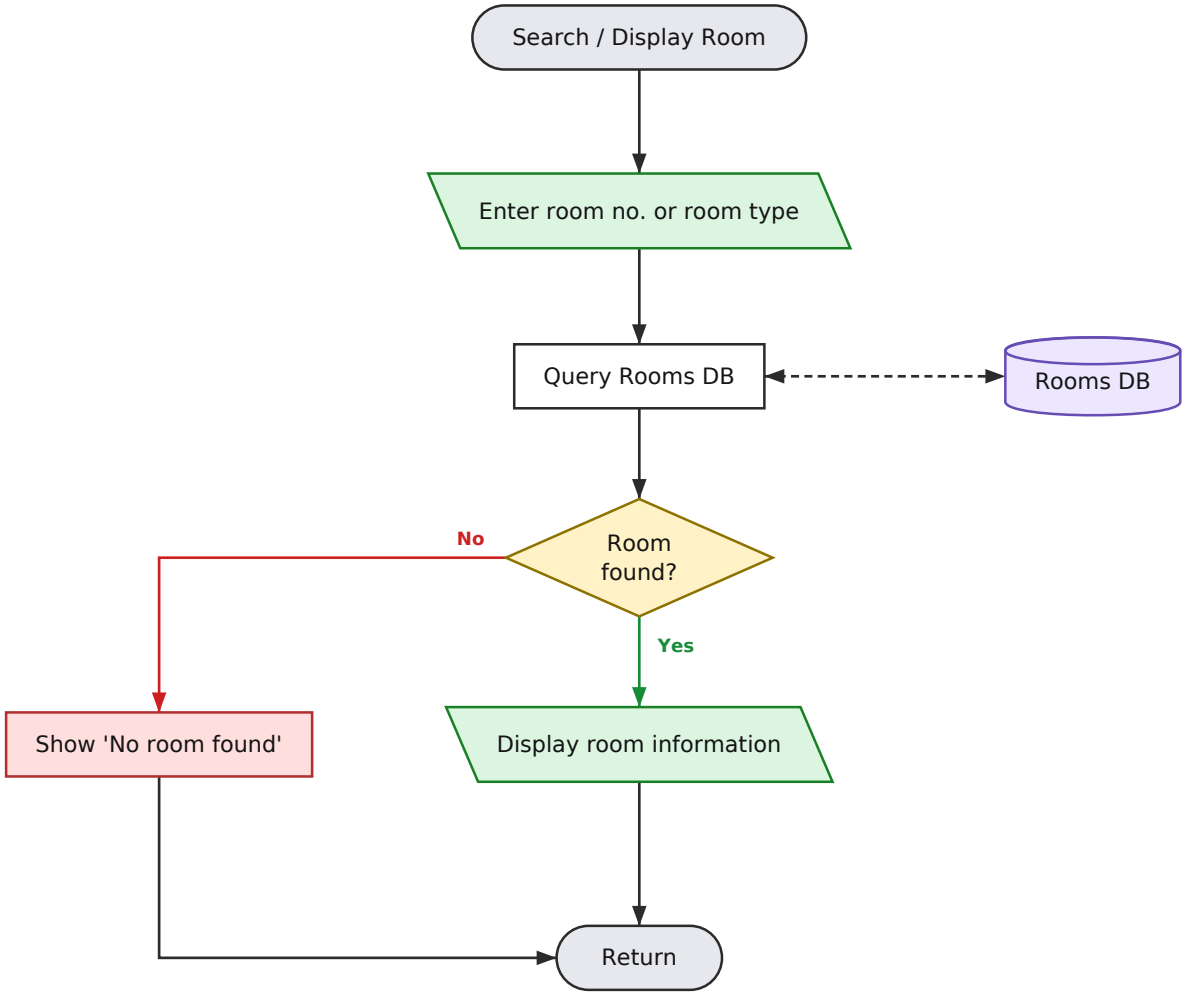


Yes No

Called from: P1. Return goes back to the calling box.

P21 Search / Display Room

Search by room number or room type.

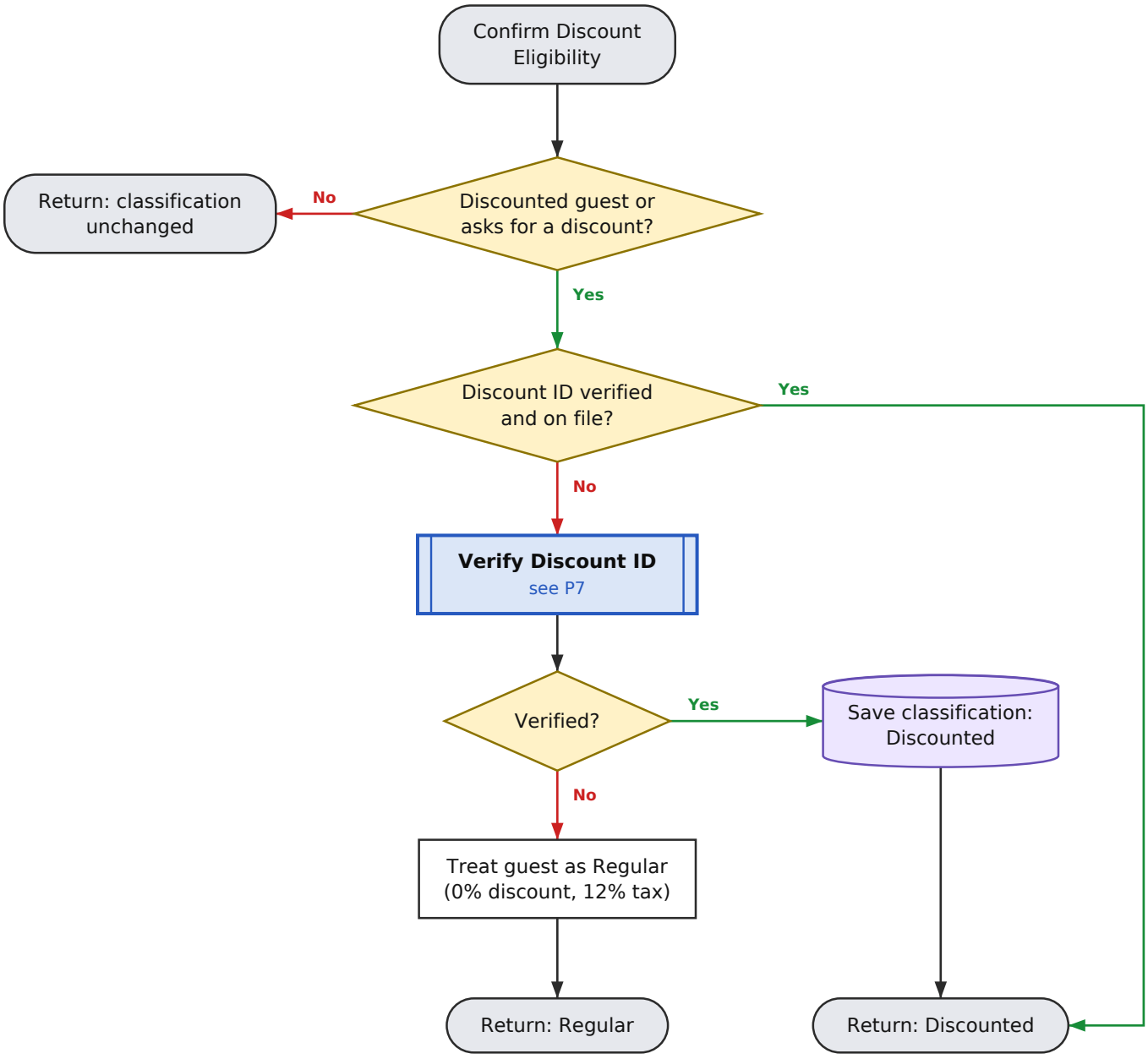


Yes No

Called from: P1. Return goes back to the calling box.

P22 Confirm Discount Eligibility

Online guests never state a classification on the website, so this is where it is settled.

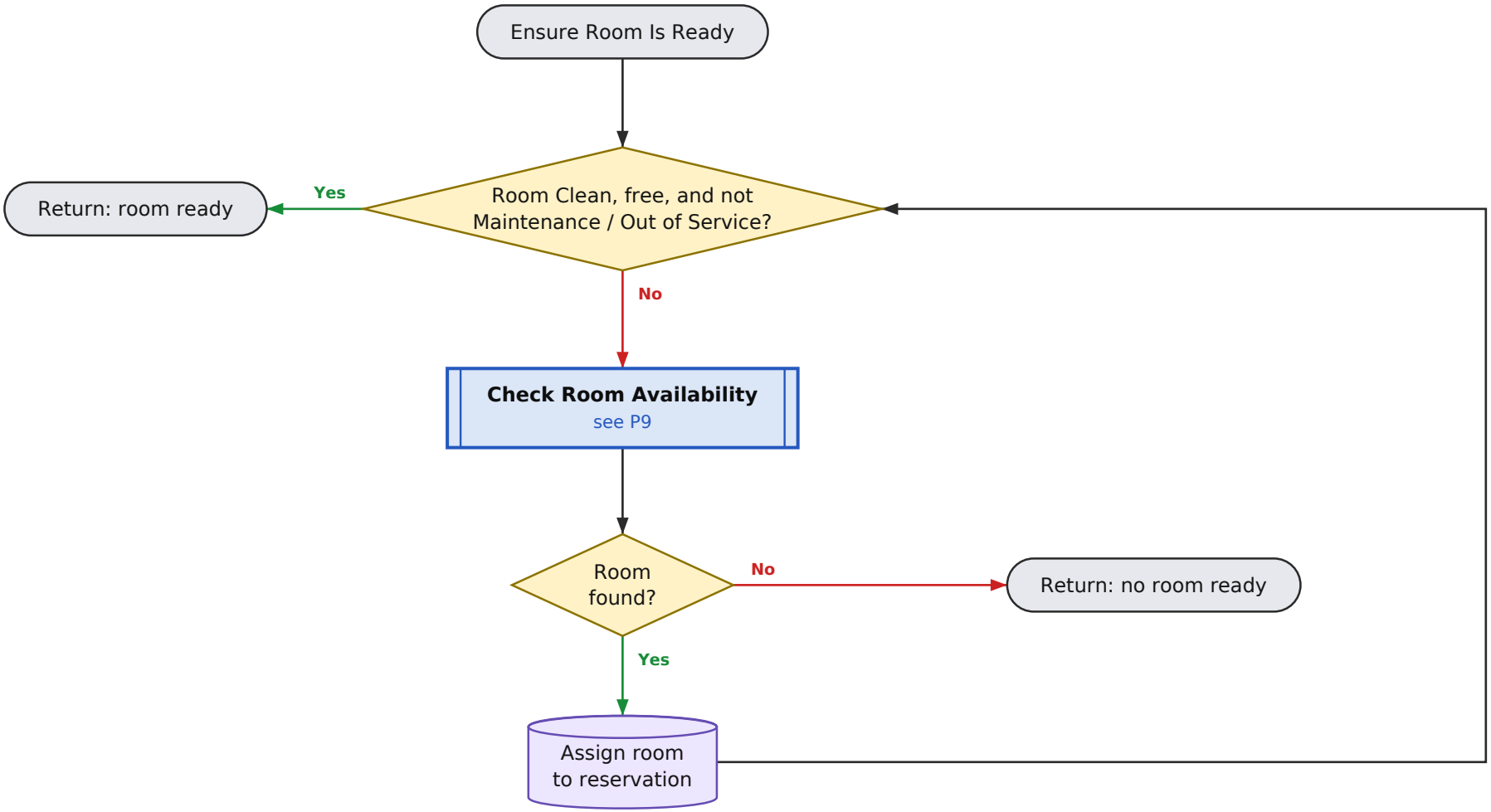


Yes No

Called from: P14. Return goes back to the calling box.

P23 Ensure Room Is Ready

A guest can only be checked in to a room that is Clean, not occupied and in service.



Yes No

Called from: P14. Return goes back to the calling box.